Differentiating Minimal-Norm Solutions to Parametric Problems with Many Solutions

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Outline

1. Introduction
2. Implicit Framework
3. At the Limit?
4. Applications

Problem Statement

Parametric / bilevel optimization

$$\min_{\theta \in \Theta} \mathcal{L}(x^\star(\theta)) \tag{Q}$$

$$\text{s.t.} \quad x^\star(\theta) \in \mathcal{S}(\theta) := \operatorname*{arg\,min}_{x \in \mathbb{R}^n} f(x, \theta). \tag{P}$$

- hyperparameter tuning [Ped16; LVD19; Ber+20]
- dictionary learning [PF11; MBP11],
- meta-learning [Fra+18; Eng+25],
- data cleaning/poisoning/... [Wan+18; Blo+21],
- model selection in statistics [Vai+15; Vai+17],
- deep learning [WK20].

First-order Methods Ideally

$$\min_{\theta \in \Theta} \mathcal{L}(x^\star(\theta))$$
$$\text{s.t.} \quad x^\star(\theta) \in \mathcal{S}(\theta) := \arg\min_{x \in \mathbb{R}^n} f(x, \theta).$$

- well-defined: $\mathcal{S}(\theta) = \{x^\star(\theta)\}$;
- for $x^\star(\theta)$: Optimality Conditions (**OC**) $\nabla_x f(x^\star(\theta), \theta) = 0$;
- $\theta^+ = \theta - \delta \nabla(\mathcal{L} \circ x^\star)(\theta)$; chain rule $\nabla_\theta(\mathcal{L} \circ x^\star) = \nabla_\theta x^\star \nabla_x \mathcal{L}$;
- IFT on the **OC**: system to solve
  $\nabla_\theta x^\star(\theta) \nabla^2_{xx} f(x^\star(\theta), \theta) + \nabla^2_{x\theta} f(x^\star(\theta), \theta) = 0$,

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Smooth Implicit Function Theorem

$$\begin{cases} F(x,y) = x^2 + y^2 - 1, \\ F(\hat{x},\hat{y}) = 0. \end{cases}$$

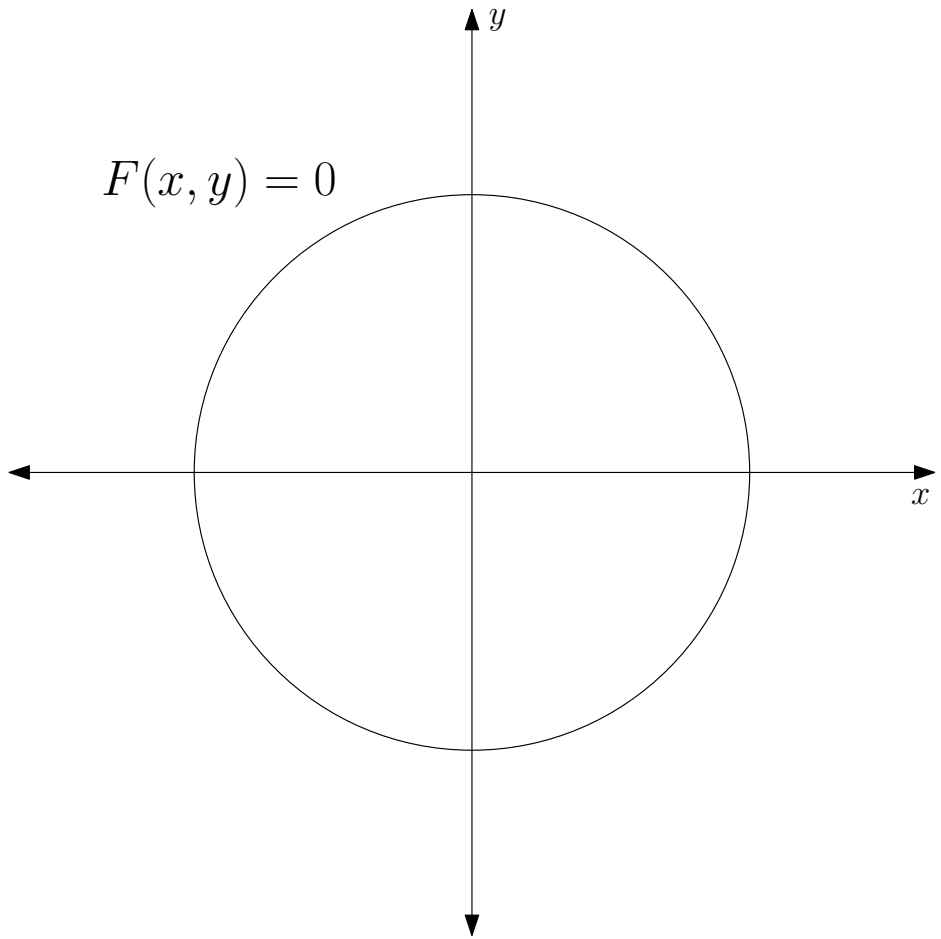
- Can we find $y = G(x)$ so that $F(x, G(x)) = 0$?

Existence

- Can we compute the gradient of G ?

Calculus

If $\frac{\partial F}{\partial y}(\hat{x},\hat{y})$ nonsingular.



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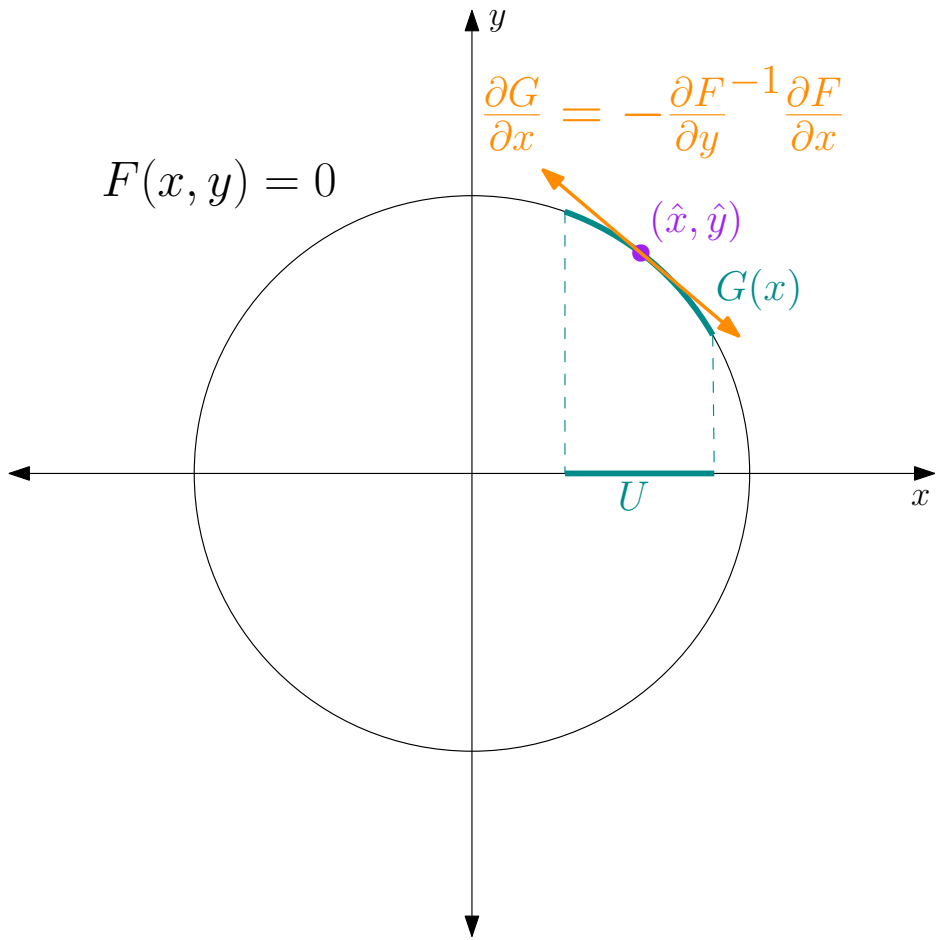
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Simple Example

Easy Least-Squares

(lower pb) $\min_x \|Ax - \theta\|_2^2/2, \quad \mathcal{N}(A) = \{0\} \Leftrightarrow A^\mathsf{T} A \succ 0$

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- Well-posedness: in general, $\mathcal{N}(A) \neq \{0\}$ so $A^\mathsf{T}A$ singular.
- Nonsmooth: some IFTs exist but no ideal derivative formula.

Regularization of Convex Problems

$$x_\varepsilon^\star(\theta) := \arg\min_x f(x, \theta) + \frac{\varepsilon}{2}\|x\|_2^2,$$
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Presentation on Least-Squares

(lower pb)	$\min_x \|Ax-\theta\|_2^2/2 + \frac{\varepsilon}{2}\|x\|_2^2$
solution	$x_\varepsilon^\star(\theta) = (A^\mathsf{T}A + \varepsilon I)^{-1}A^\mathsf{T}\theta,$
Jacobian	$J_{x_\varepsilon^\star}(\theta) = (A^\mathsf{T}A + \varepsilon I)^{-1}A^\mathsf{T},$
unregularized solutions	$\mathcal{S}(\theta) = A^\dagger\theta + \mathcal{N}(A), x_{\min}^\star(\theta) = A^\dagger\theta$
unregularized Jacobian	$J_{x_{\min}^\star}(\theta) = A^\dagger$

Problematic

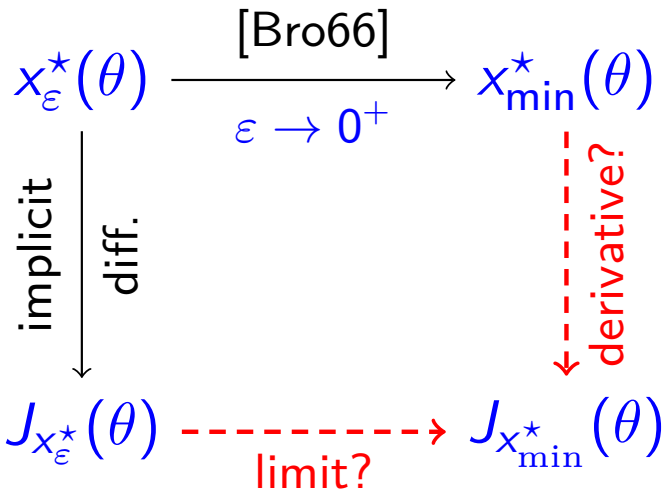
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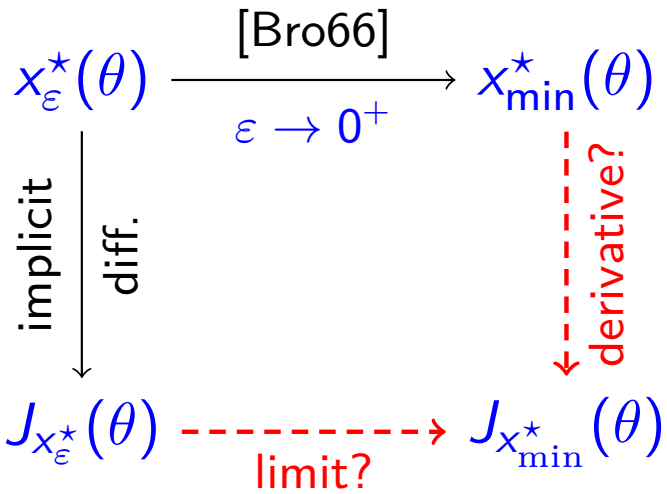
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Huber/LASSO/... not $\mathcal{C}^2$. All nonsmoothness hidden under **conservative theory**: valid nonsmooth IFT, multiple elements.

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Pseudoinverse Option

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- **pseudoinverse for quadratic costs** [Vic+22],
- ~~convex~~ case: parametric Morse-Bott [AM22; Bol+26],
- ~~convex~~ case: Polyak-Łojasiewicz assumption, select then differentiate [Mas+26],
- value function [SV26].

Framework

$$x_\varepsilon^\star(\theta) : 0 = \nabla_x f(x_\varepsilon^\star(\theta), \theta) + \varepsilon x_\varepsilon^\star(\theta) = F_\varepsilon(x_\varepsilon^\star(\theta), \theta).$$

Conservative (Nonsmooth) IFT Step

$$J_{F_\varepsilon}(x, \theta) = \left\{ (H + \varepsilon I \ M) : (H \ M) \in J_{\nabla_x f}(x, \theta) \right\}, \quad \text{at } (x_\varepsilon^\star(\theta), \theta)$$

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Convergence?

$$\lim_{\varepsilon \to 0} -(H_\varepsilon + \varepsilon I)^{-1} M_\varepsilon \overset{?}{=} -H_0^\dagger M_0 \overset{?}{\in} J_{x_{\min}^\star}(\theta)$$

Convergence in our Setting (see Schechtman [Sch26])

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- We need uniform boundedness in ε for the limit.
- Nonsmoothness: technicalities ($\rightsquigarrow$ valid for all elements)

uniform boundedness: $\forall\ K$ compact, $\exists\ C_K > 0, \varepsilon_K > 0$ s.t.

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Relaxing Nonsingularity

Goal: bounded derivatives without assuming nonsingularity.

Idea: uniform range assumption (R)

$$(M_\varepsilon \in \operatorname{range}(H_\varepsilon)) \quad M_\varepsilon = H_\varepsilon W_\varepsilon, \quad \|W_\varepsilon\| \leqslant C$$

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Nonsingularity: $\forall\ (H_0\ M_0) \in J_F(x^\star_{\min}(\theta), \theta), \det(H_0) \neq 0$ [Bol+21].
Continuity of det: nonsingularity $\Rightarrow$ uniform range; **(R)** can fail.

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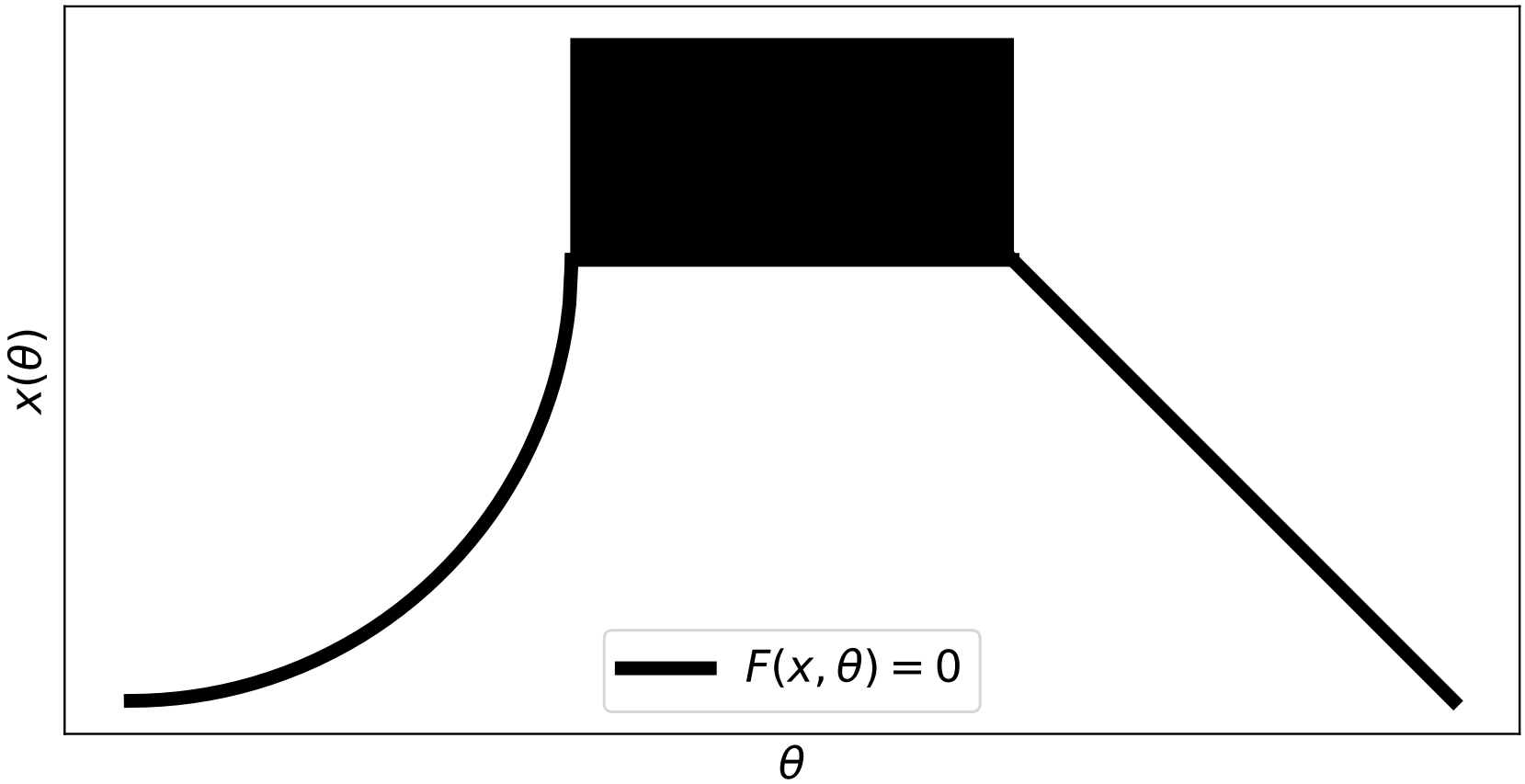
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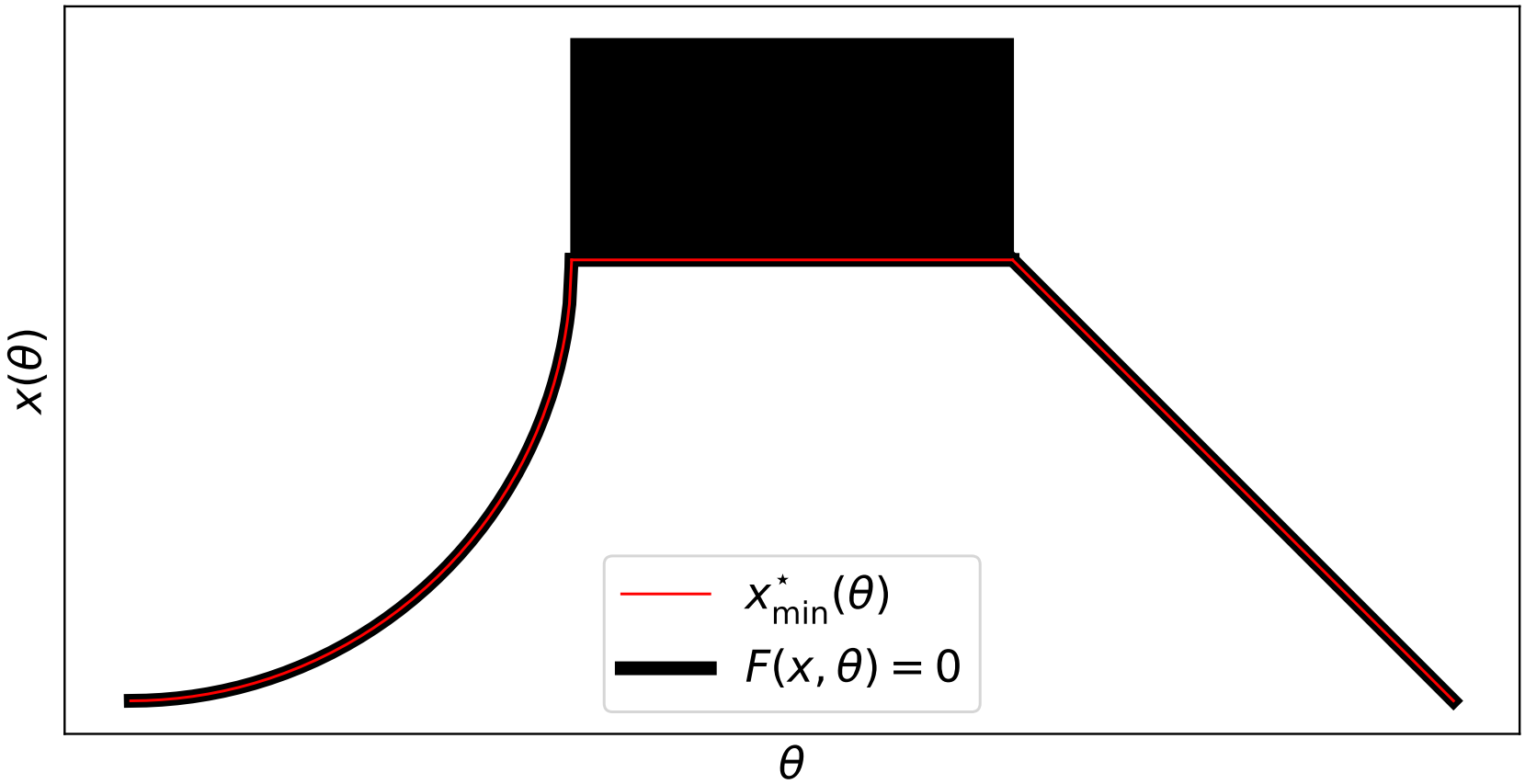
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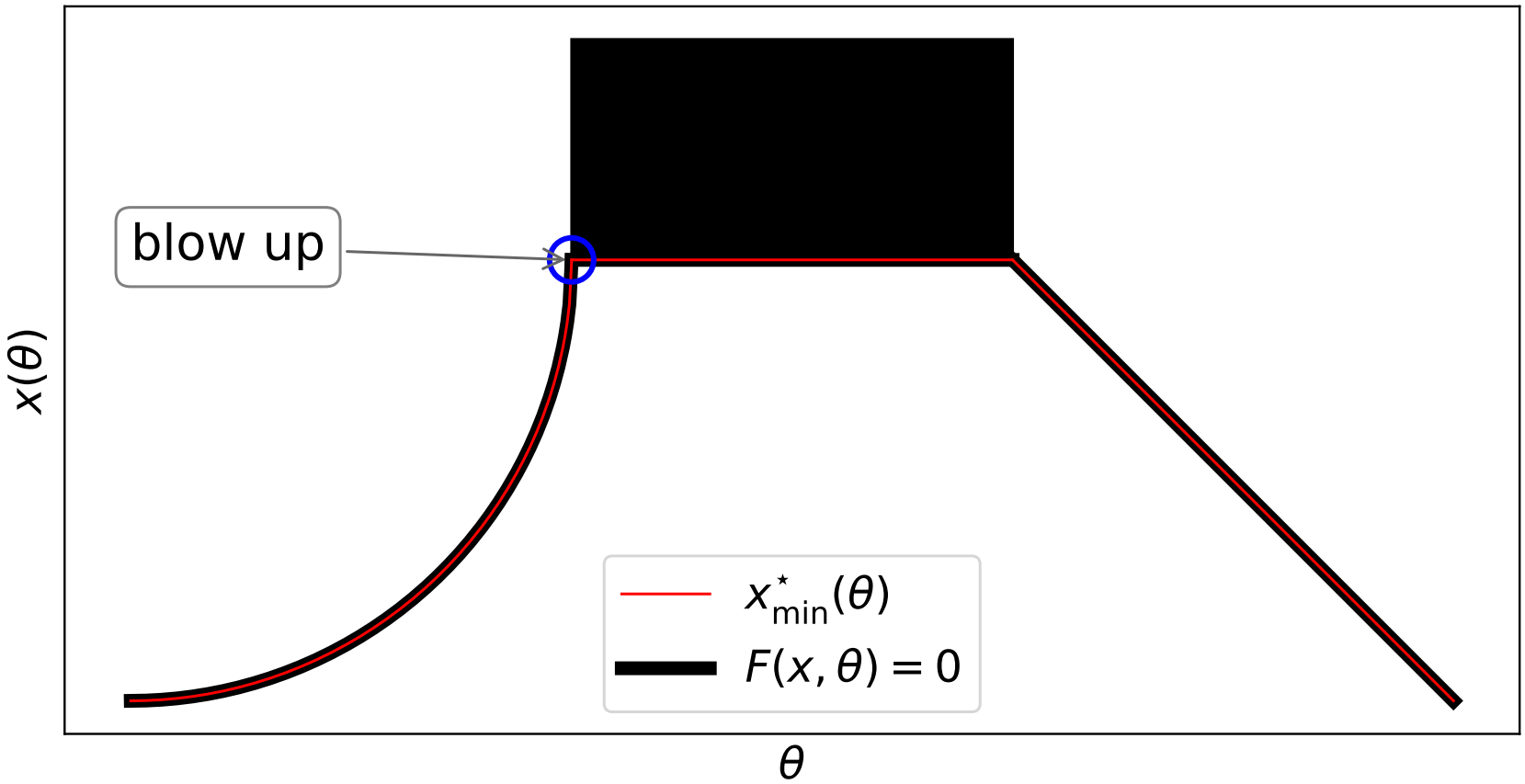
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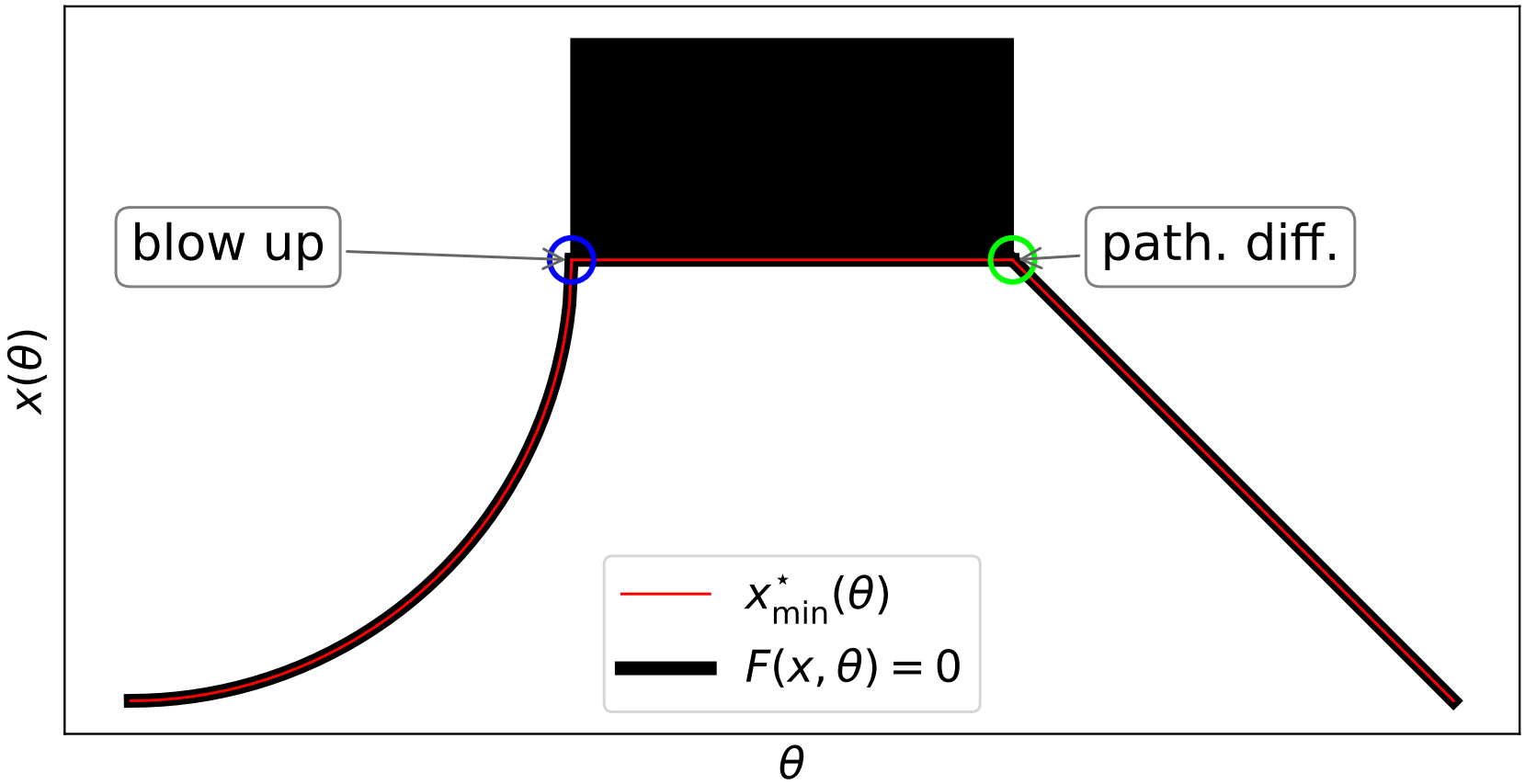
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About limits

- Boundedness ⇒ converging subsequences.
- Pseudoinverse can fail (independently of nonsmoothness):
- there can be a calculation mismatch.

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Actual Limits vs Pseudoinverse Formula

Let $0 < q < 1 < p$, $c, c' > 0$, and $H_\varepsilon = \mathrm{diag}(0, \varepsilon^p, c\varepsilon, \varepsilon^q, c')$,

[with range] $\quad (H_\varepsilon + \varepsilon I_n)^{-1} H_\varepsilon \to \mathrm{diag}(0, 0, c/(1+c), 1, 1)$

$$\neq \mathrm{diag}(0, 0, 0, 0, 1) = H_0^\dagger H_0.$$

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Observations:

- eigenvalues $\equiv 0$ (rank deficiency) or $\equiv c'$ ($\nrightarrow 0$) ✓;
- otherwise, curvature *vanishes* with ε;
- if $\lambda_i(H_\varepsilon) = o(\varepsilon)$ "fast", equals pseudoinverse,
- if $\varepsilon = o(\lambda_i(H_\varepsilon))$ "slow", mismatch,
- if $\lambda_i(H_\varepsilon) \sim \varepsilon$ "critical", smaller mismatch.

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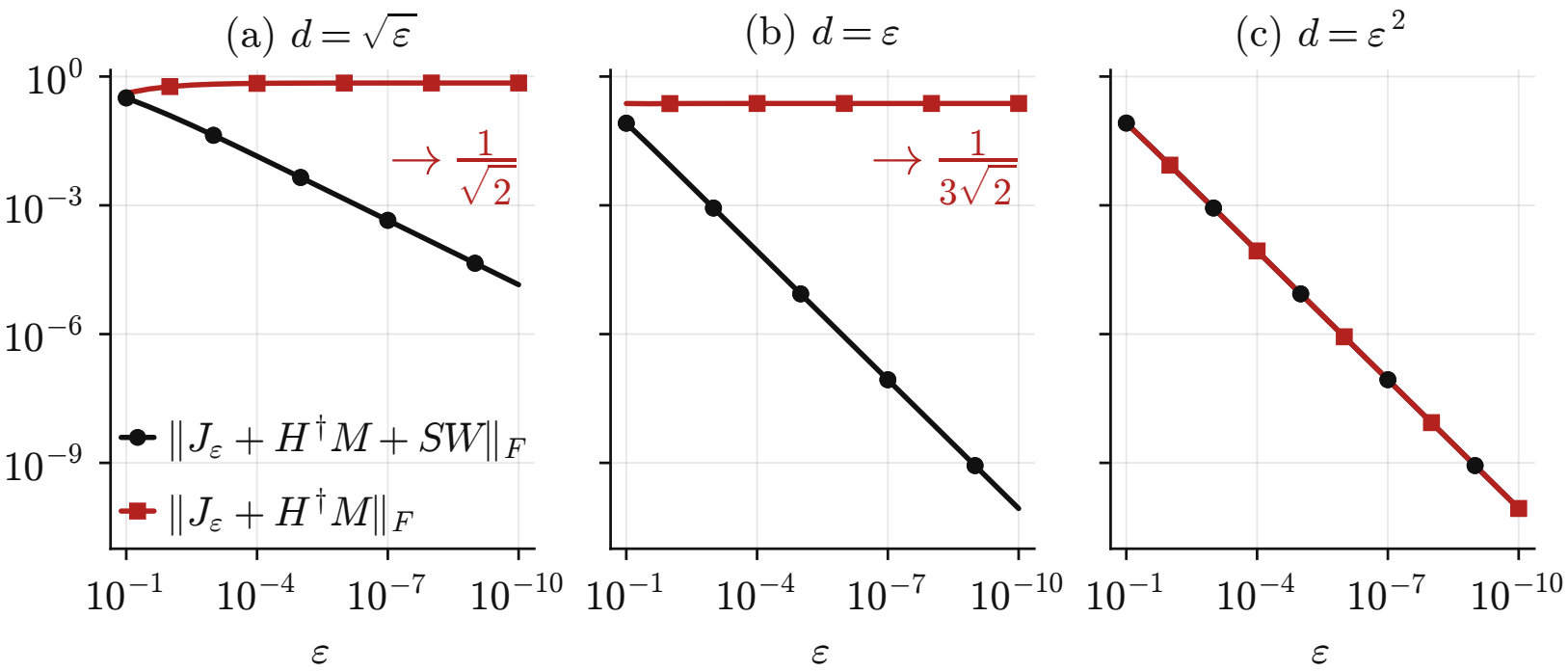
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Huber Regression and LASSO (lower level)

$$\min_{x\in\mathbb{R}^n} \sum_{i=1}^{n_{\text{lower}}} h_\delta((Ax-\theta)_i), h_\delta(r) = \begin{cases} \frac{1}{2}r^2, & |r| \leqslant \delta, \\ \delta|r| - \frac{1}{2}\delta^2, & |r| > \delta, \end{cases}, \delta > 0$$

$$\text{or} \quad \min_{x\in\mathbb{R}^n} \frac{1}{2}\|Ax-\theta\|_2^2 + e^\lambda \|x\|_1.$$

About the uniform bound:

Good news: global bound for the regularized Jacobians ✓
Bad news: this proof is specific for Huber/LASSO.

- Numerically, finite number of $\varepsilon > 0$;
- compute $J_{x_\varepsilon^\star}$, verify if they "stabilize" when $\varepsilon \to 0$
- compute the limit $\varepsilon = 0$ with stabilized sequence

Huber Regression and LASSO (lower level)

$$\min_{x\in\mathbb{R}^n} \sum_{i=1}^{n_{\text{lower}}} h_\delta((Ax-\theta)_i), h_\delta(r) = \begin{cases} \frac{1}{2}r^2, & |r| \leqslant \delta, \\ \delta|r| - \frac{1}{2}\delta^2, & |r| > \delta, \end{cases}, \delta > 0$$

$$\text{or} \quad \min_{x\in\mathbb{R}^n} \frac{1}{2}\|Ax-\theta\|_2^2 + e^\lambda \|x\|_1.$$

About the uniform bound:

Good news: global bound for the regularized Jacobians ✓
Bad news: this proof is specific for Huber/LASSO.

- Numerically, finite number of $\varepsilon > 0$;
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Illustration: Huber Experiment

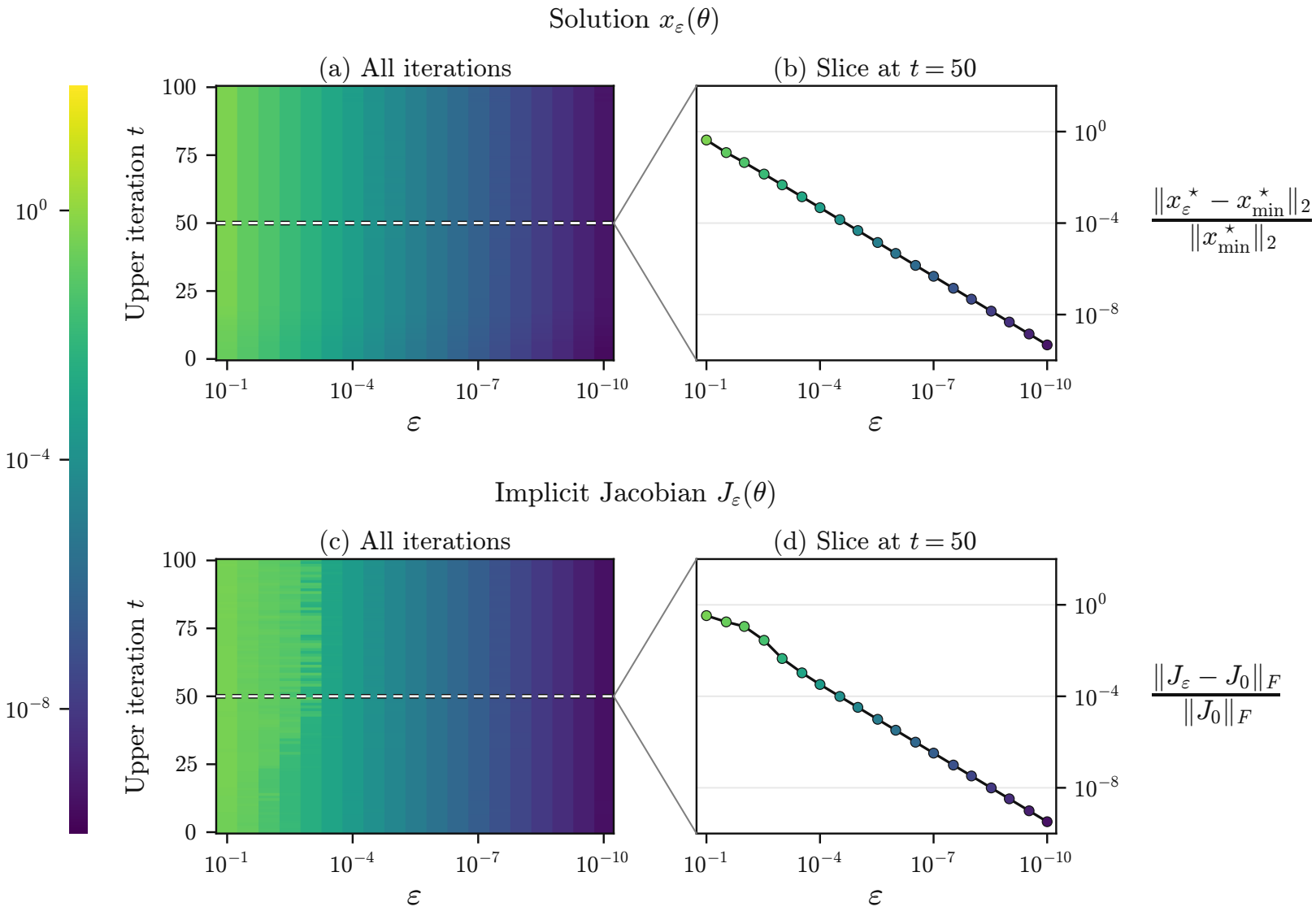
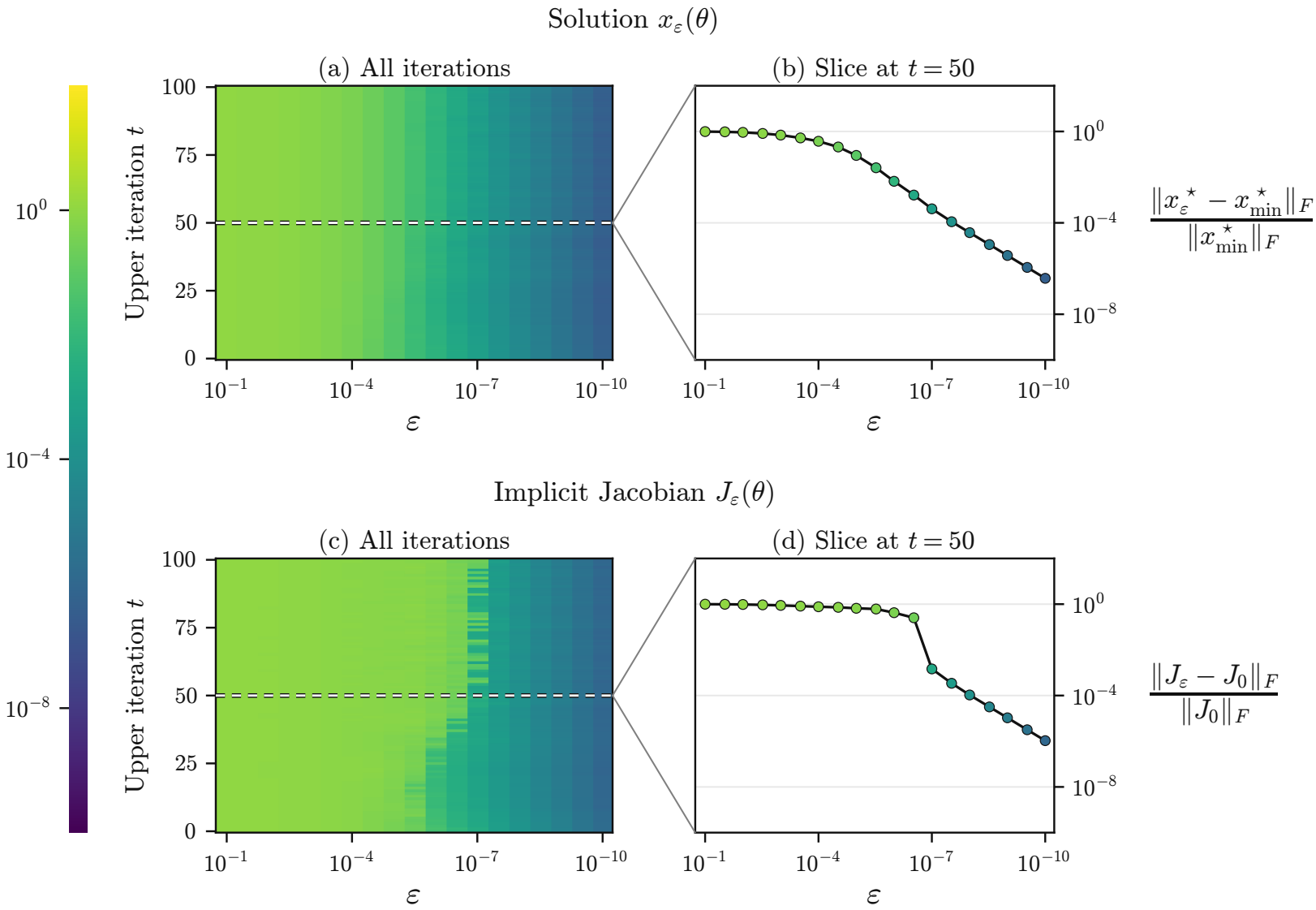


Illustration: LASSO Experiment



Conclusion

Summary

- theoretical justification of relevant practical methods;
- applicable to Huber & LASSO;
- some convincing numerical evidence.

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Uncharted directions

- assumptions may be hard to verify;
- ε-regularization limited by constraints; see also [MO24];
- deeper understanding of numerical behavior.

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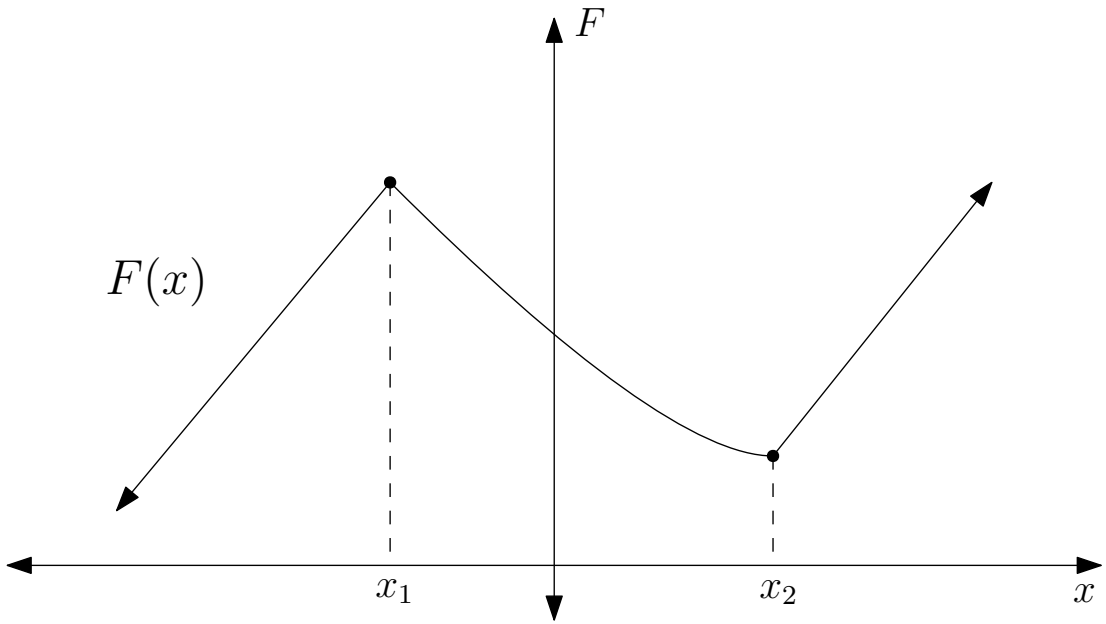
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Illustration: captions

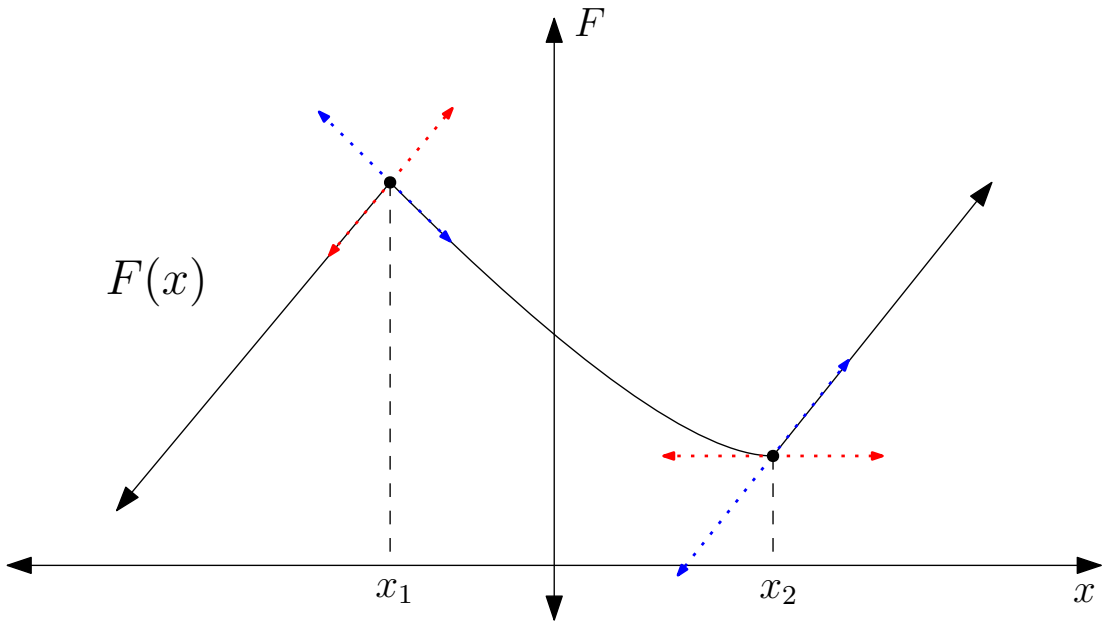
- Huber: *Top*, convergence of $x_\varepsilon^\star(\theta)$. *Bottom*, error on $J_{x_\varepsilon^\star}(\theta)$. Abscissa: ε; ordinate: upper iteration. Heatmaps: $\text{errors}(\varepsilon, t)$ ($\simeq$ independent of t). Curves: iteration $t = 50$. Linear regime is observed for $x_\varepsilon^\star$ all along, observed for $J_{x_\varepsilon^\star}$ with smaller ε.
- LASSO: *Top*, convergence of $x_\varepsilon^\star(\theta)$. *Bottom*, error on $J_{x_\varepsilon^\star}(\theta)$. Abscissa: ε; ordinate: upper iteration. Heatmaps: $\text{errors}(\varepsilon, t)$ ($\simeq$ independent of t). Curves: iteration $t = 50$. Linear regime observed for $x_\varepsilon^\star$ and $J_{x_\varepsilon^\star}$ at lower ε – support identification?

Nonsmooth but Locally Lipschitz Functions



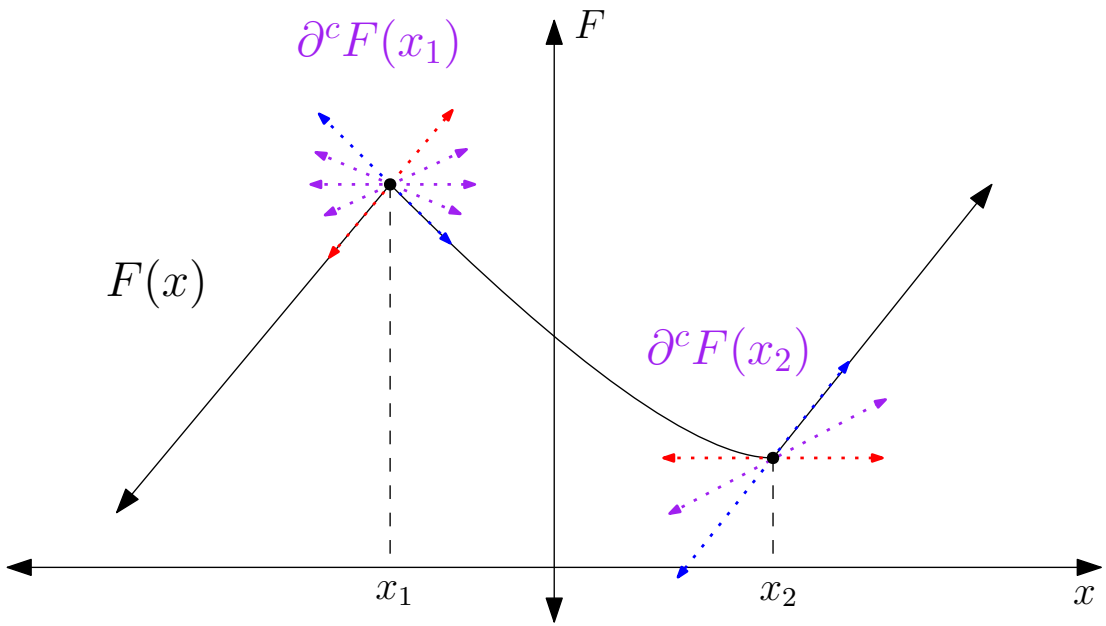
$$\partial^c F(x) := \operatorname{conv}\left(\{V : F'(x_k) \to V \text{ for } \{x_k\}_k \to x, F'(x_k) \text{ exists } \}\right)$$

Nonsmooth but Locally Lipschitz Functions



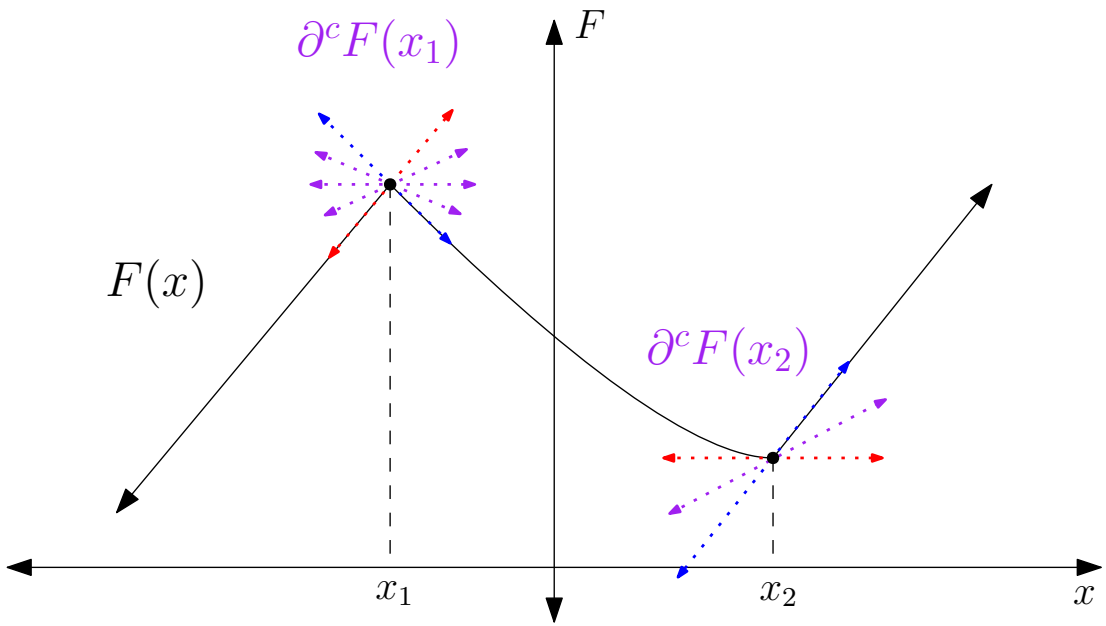
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- IFT **Existence**: theorems [Cla90; Rob91]...
- IFT **Calculus**: some issues.

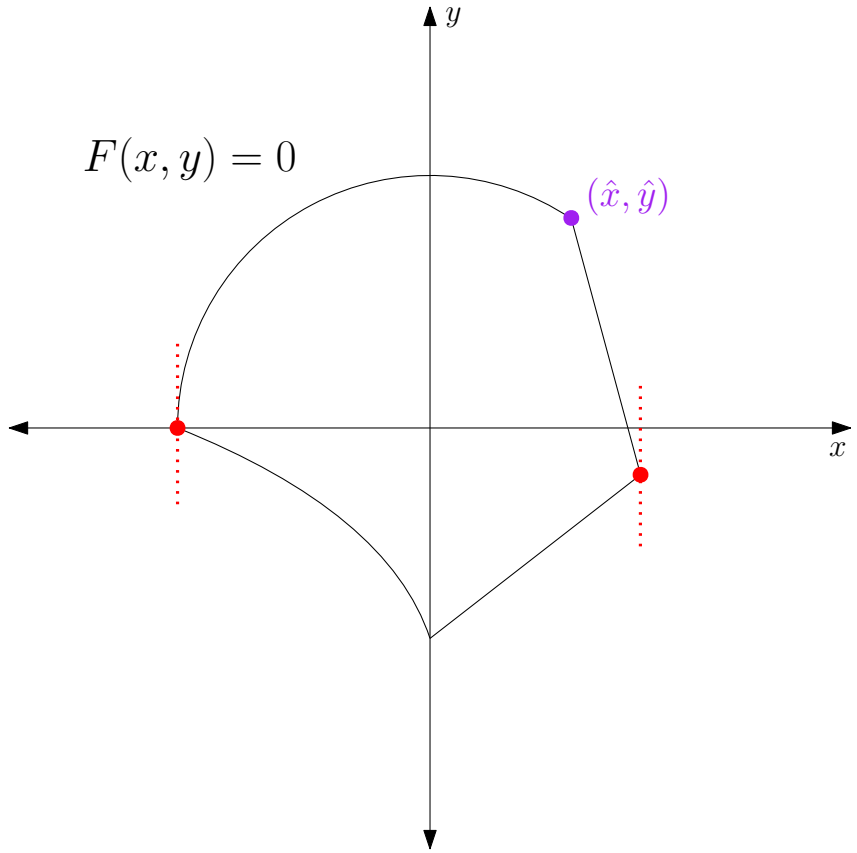
Nonsmooth IFT with Clarke Differential

Theorem ([Cla90]...)

Let $F : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^m$ *be locally Lipschitz and* $F(\hat{x}, \hat{y}) = 0$.

If, $\forall (A\ B) \in \partial^c F(\hat{x}, \hat{y}), B$ *is invertible, then* $\exists\ U \subseteq \mathbb{R}^n$ *a neighborhood of* $\hat{x}$ *and a locally Lipschitz function* $G(x)$ *so that*

$$F(x, G(x)) = 0 \qquad \forall x \in U.$$



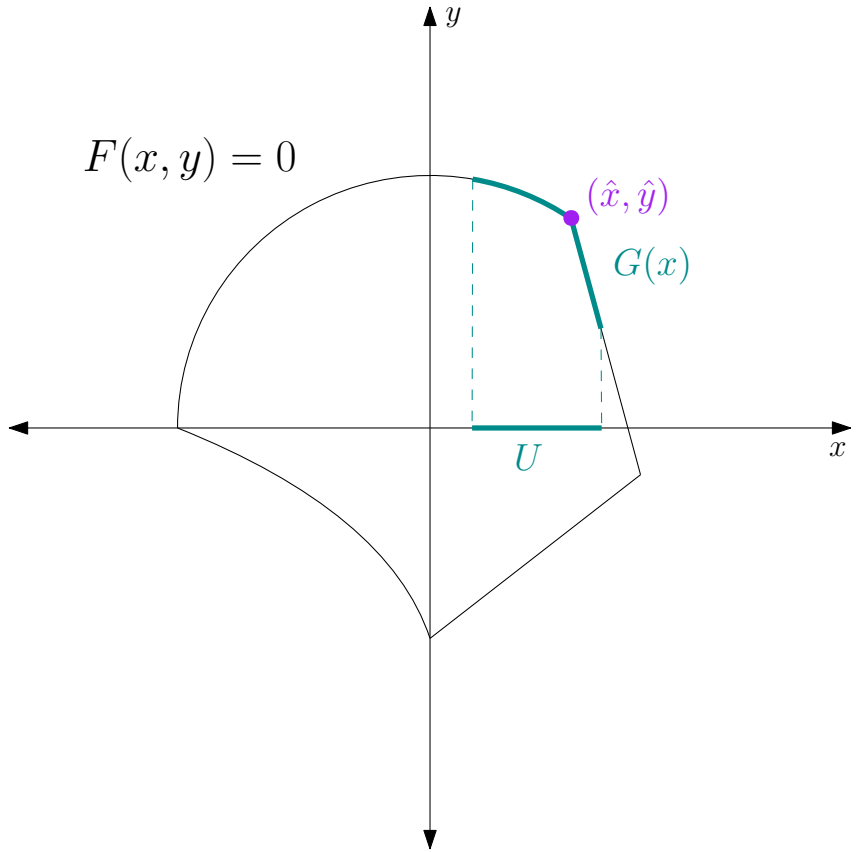
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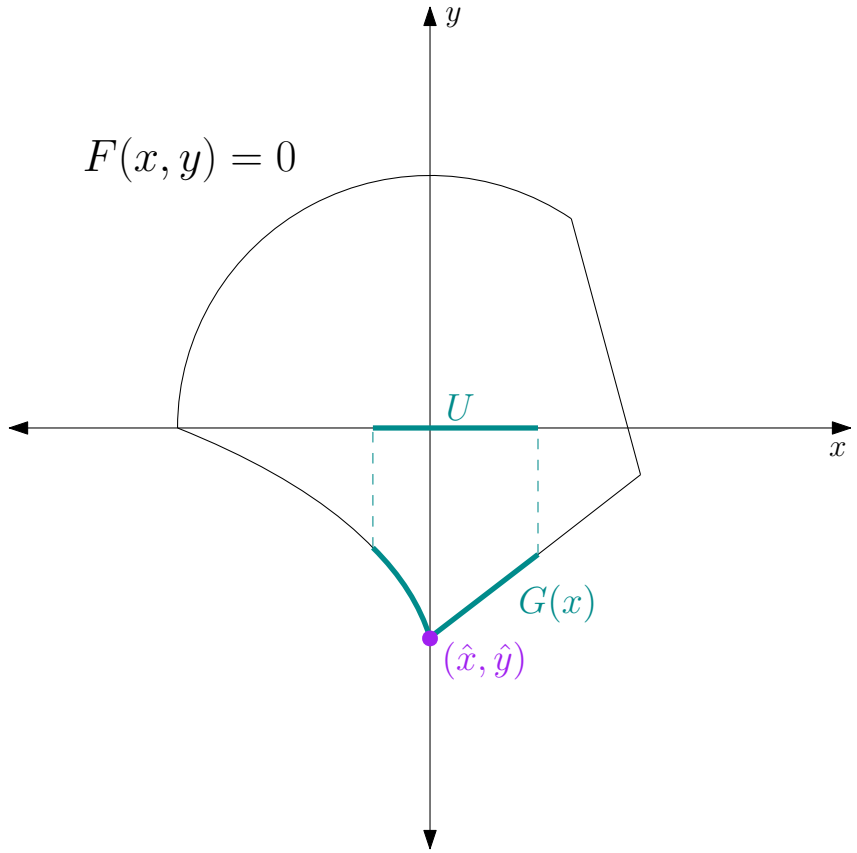
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But by conv: many B elements.

Conservative Fields (/Mappings/Jacobians)

Definition (conservative field of a locally Lipschitz function [BP20])

Let $F : \mathbb{R}^n \to \mathbb{R}$, $D_F : \mathbb{R}^n \rightrightarrows \mathbb{R}$ is a conservative field of F if

- $\forall\, x, D_F(x) \neq \varnothing$
- D_F is locally bounded and has closed graph;
- $\forall\, \gamma : [0,1] \to \mathbb{R}^n$ absolutely continuous ($\dot{\gamma}(t)$ exists a.e. t),

$$\underset{\text{rule}}{\text{chain}} \qquad \frac{\mathrm{d}}{\mathrm{d}t} F(\gamma(t)) = \langle u, \dot{\gamma}(t) \rangle, \quad \forall\, u \in D_F(\gamma(t)), \text{a.e. } t$$

Chain rule justifies backpropagation (Jax, Pytorch...)

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Some Properties of Conservative Fields

- Generalization of ∂^c but not a unique D_F.
- Usual calculus rules: sum, product, **composition**.
- Almost everywhere, $D_F(x) = \nabla F(x)$ (see [Rad19]).
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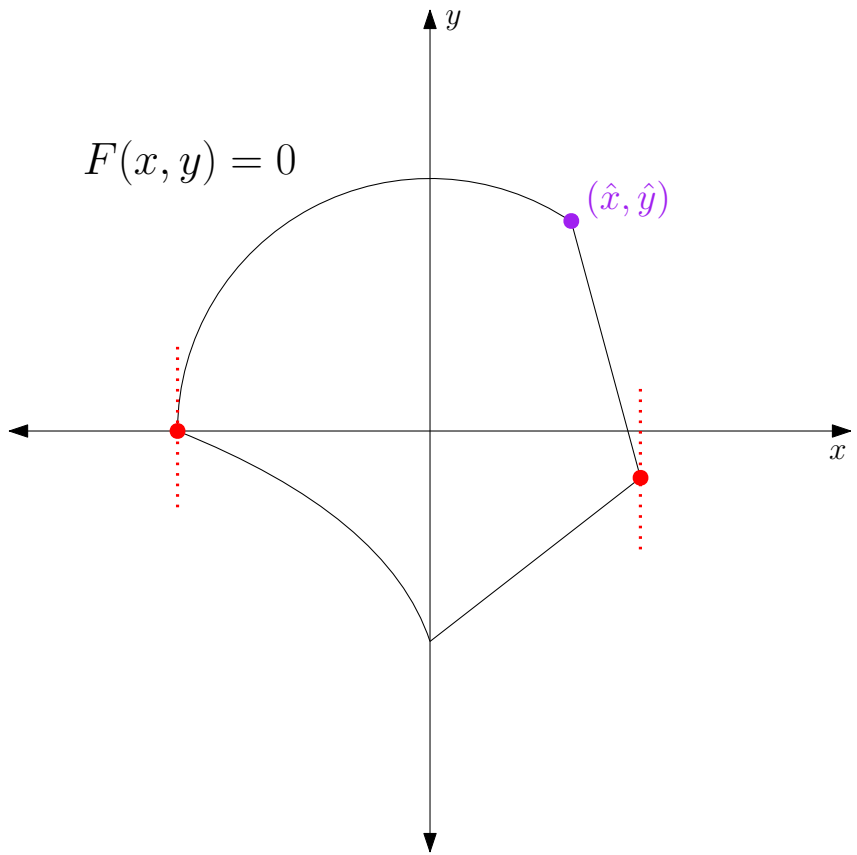
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Theorem ([Bol+21])

Let $F : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^m$ *be path diff. and* $F(\hat{x}, \hat{y}) = 0$.
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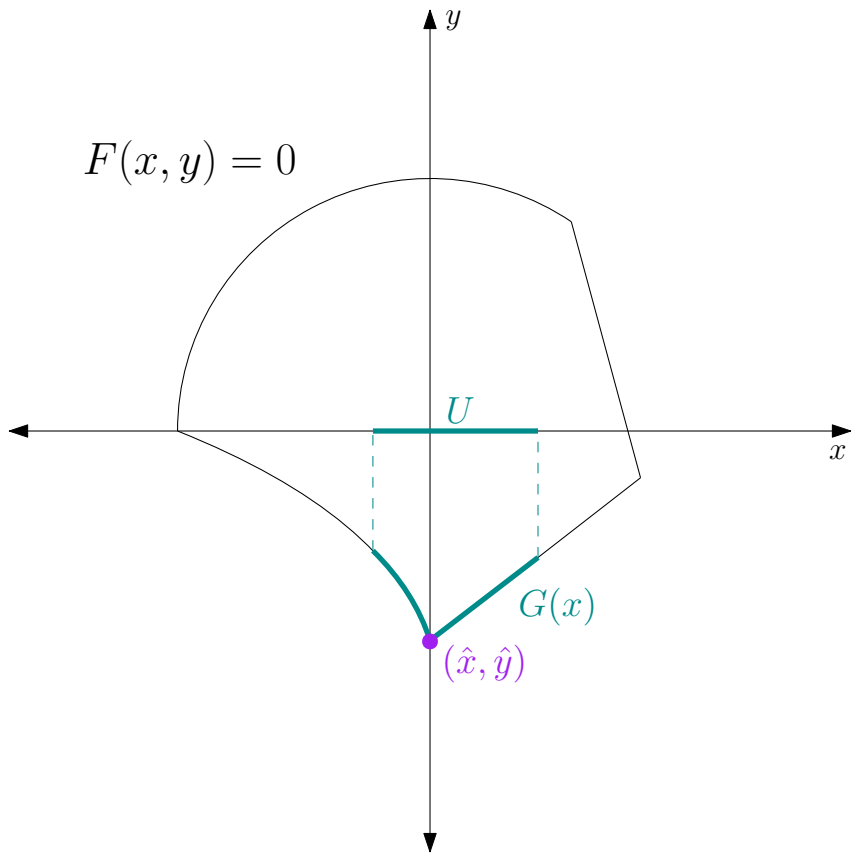


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$$\forall x \in U, \quad F(x, G(x)) = 0,$$

$$D_G(x) = \left\{ \begin{array}{l} -B^{-1}A : (A\ B) \\ \in D_F(x, G(x)) \end{array} \right\}.$$

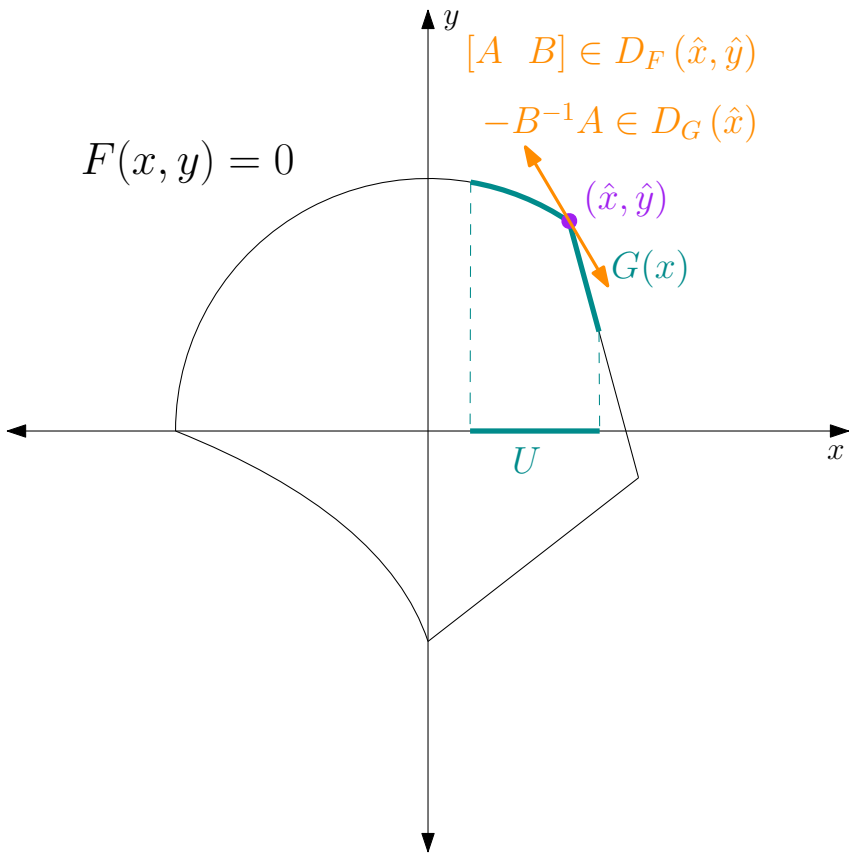
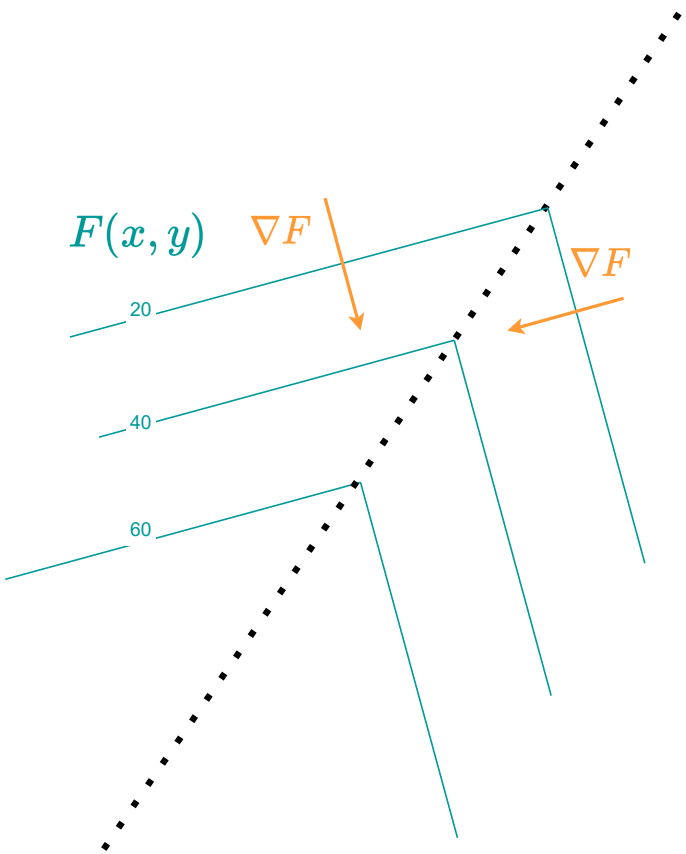
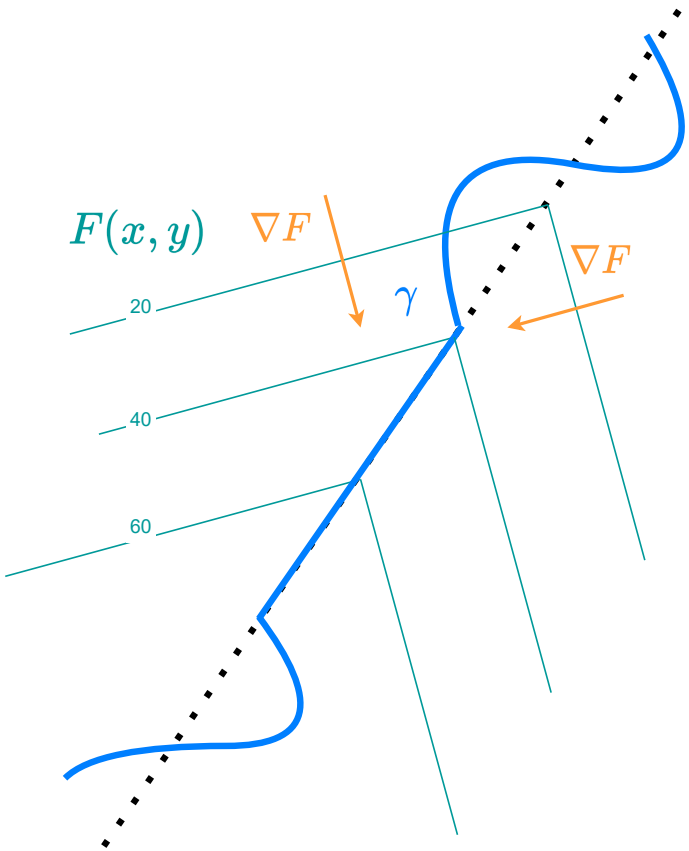


Illustration on Piecewise Affine Functions



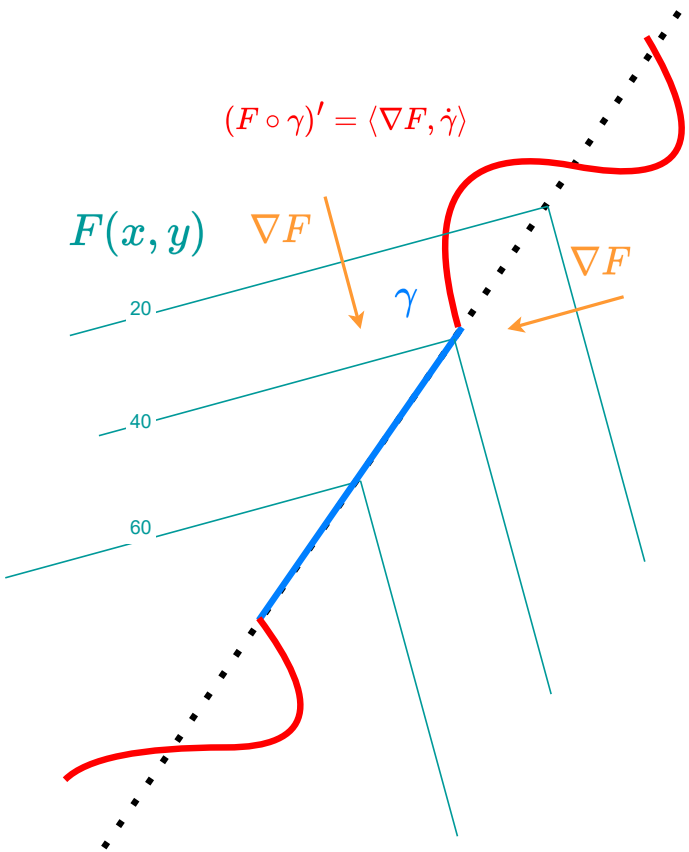
Locally Lipschitz function $F : \mathbb{R}^2 \to \mathbb{R}$ with piecewise constant gradient and kink in black.

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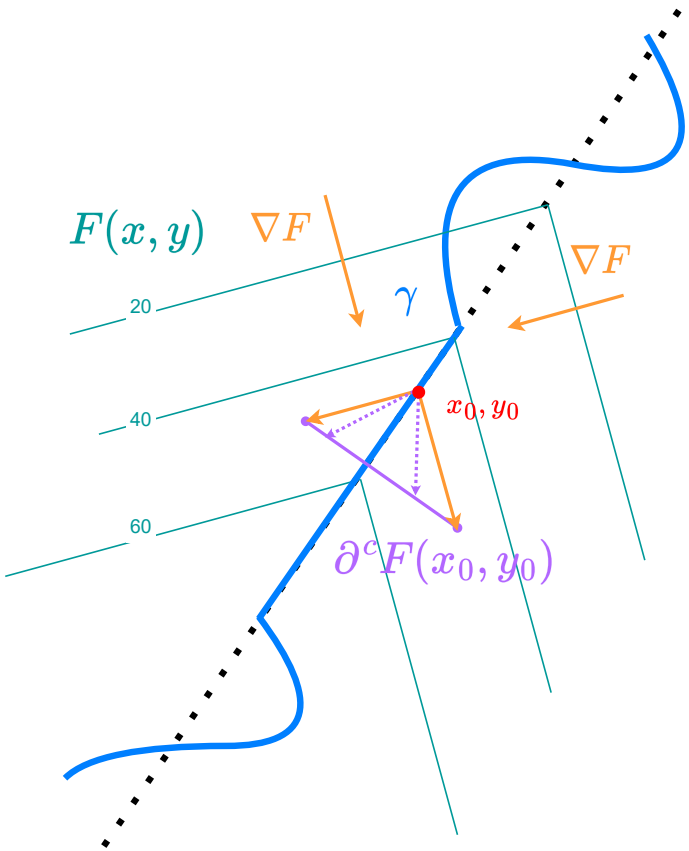
Consider some absolutely continuous curve $\gamma\colon [0,1] \to \mathbb{R}^2$.

Illustration on Piecewise Affine Functions



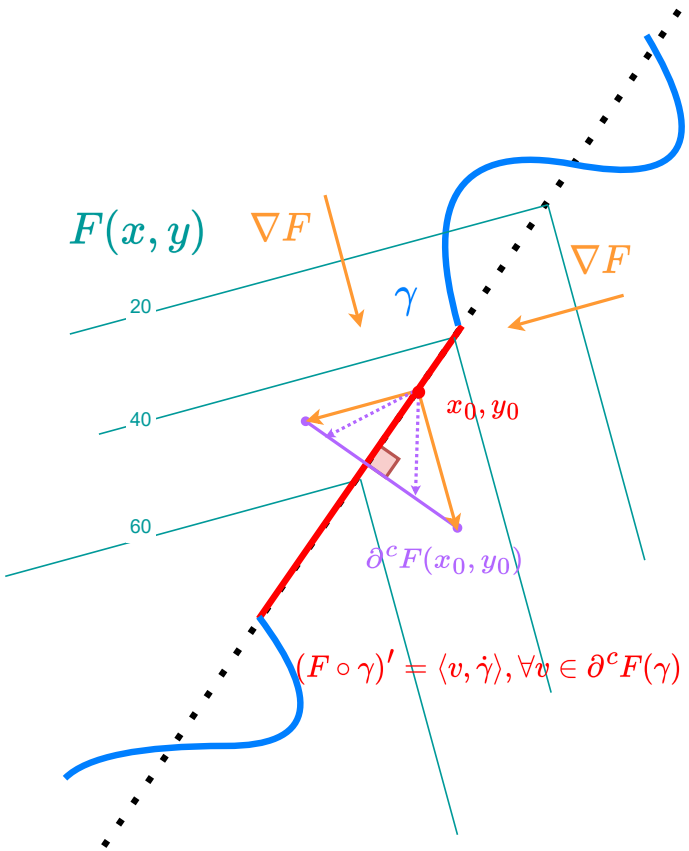
The red parts of $\gamma([0, 1])$ only see the bad region for a set of measure zero in t.

Illustration on Piecewise Affine Functions



For a point in the bad region, the Clarke subdifferential has this form.

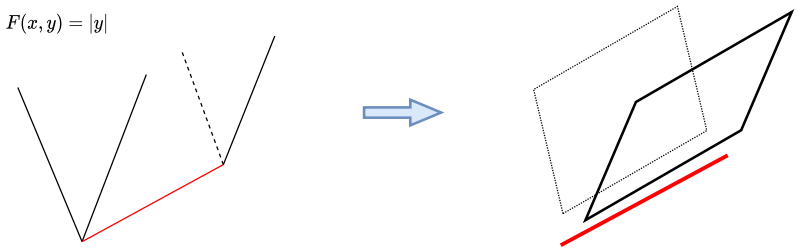
Illustration on Piecewise Affine Functions



The formula holds for any point in the bad region.

Relation with Definability and o-minimal Geometry

Piecewise affine functions are easily decomposed as earlier

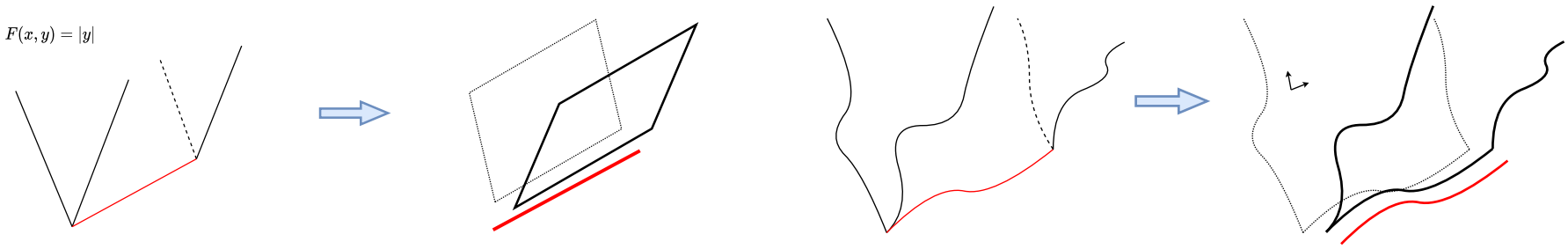


Definability in our setting

- definable geometry generalizes this beyond affine functions;
- graphs of definable functions are decomposable into manifolds that are locally like piecewise affine ("Whitney stratification");
- locally Lipschitz definable $\Rightarrow$ path-differentiable;
- we will assume "everything" is definable.

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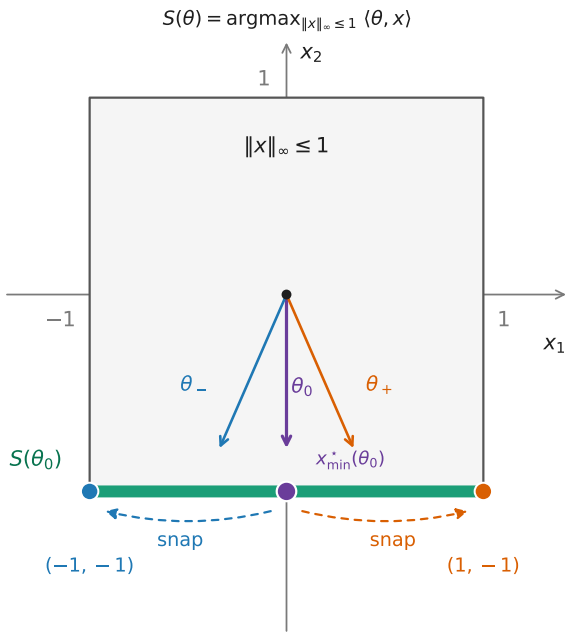


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Failure for constrained/composite problems

Consider: $\max \theta^\mathsf{T} x, \|x\|_\infty \leqslant 1 \Leftrightarrow \max \theta^\mathsf{T} x - \iota_{B_\infty(0,1)}(x)$.



We have $x^\star_{\min}(\theta) = \operatorname{sgn}(\theta)$ ($\operatorname{sgn}(0) = 0$). For $\theta_0 = (0, -1)$, $\mathcal{S}(\theta_0)$ is in green and $x^\star_{\min}(\theta_0) = (0, -1)$. There is no continuity thus no path-differentiability. Perturbing θ_0 collapses the edge $\mathcal{S}(\theta_0)$ to a corner.

Illustration: Different Regimes (1)

Test problem: duplicated soft-thresholding (trivial LASSO)

$$\min_m h(m) := \frac{1}{2}\|m-\theta\|_2^2 + \alpha\|m\|_1 \tag{1}$$

Version with duplicate variable ($m = x_1 + x_2$):

$$\min_{x_1,x_2} \tilde{h}(x_1,x_2) := \frac{1}{2}\|x_1+x_2-\theta\|_2^2 + \alpha(\|x_1\|_1 + \|x_2\|_1) \tag{2}$$

with ε regularization; we take $\theta \in \mathbb{R}^n_+$ and $\alpha = \theta_k$.

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with ε regularization; we take $\theta \in \mathbb{R}^n_+$ and $\alpha = \theta_k$.

$\alpha = \theta_k \Rightarrow$ nonsmoothness on index k; $J^1_{i\neq k} = J^2_{i\neq k}$, $J^1_k \neq J^2_k$.
We **intentionally** construct a mismatch on coordinate k.

Detailed range assumption

Uniform range assumption **(R)**

Let $O \subseteq \Theta$ open, $\exists\ \bar{\varepsilon} > 0$ s.t. $\forall\ K \subseteq O$ compact, $\exists\ C_K > 0$, s.t. $\forall\ \theta \in K, \forall\ 0 < \varepsilon \leqslant \bar{\varepsilon}$ and $\forall\ (H_\varepsilon\ M_\varepsilon) \in J_F(x_\varepsilon^\star(\theta), \theta)$, $H_\varepsilon \succeq 0$ and $M_\varepsilon = H_\varepsilon W_\varepsilon$ with $\|W_\varepsilon\| \leqslant C$:

$$-\left(H_\varepsilon + \varepsilon I_n\right)^{-1} M_\varepsilon = -\underbrace{\left(H_\varepsilon + \varepsilon I_n\right)^{-1} H_\varepsilon}_{0 \preceq \cdot \preceq I}\ \underbrace{W_\varepsilon}_{\|\cdot\| \leqslant C}\ .$$

Vanishing curvature

Counterexample (in 1D)

$$f(x,\theta) = \psi(x-\theta), \psi(t) = \begin{cases} t^4/4 & |t| \leqslant 1, \\ |t| - 3/4 & |t| > 1. \end{cases}$$

- $x(\theta) = \theta, J_{x^\star_{\min}} \equiv 1 \neq -H_0^\dagger M_0 = -0^\dagger 0 = 0$;
- $H_\varepsilon \sim \varepsilon^{2/3}, M_\varepsilon = -H_\varepsilon$: $\lim_{\varepsilon \to 0} (\varepsilon^{2/3} + \varepsilon)^{-1} \varepsilon^{2/3} = 1$.

Assumptions to use the pseudoinverse

Eigenvalues of Hessian blocks must not vanish or vanish fast.

Statement for the pseudoinverse formula

- **(G)** $\exists\ \lambda^*, \forall\ \varepsilon \leqslant \varepsilon_G, \mathrm{spectrum}(H_\varepsilon) \subseteq \{0\} \cup [\lambda^*; +\infty[$ ("gap")
- **(V)** for all converging sequences H_ε with ordered eigenvalues $\lambda_1(H_\varepsilon) \geqslant \ldots$, $\lambda_i(H_\varepsilon) \to 0 \Rightarrow \lambda_i(H_\varepsilon) = o(\varepsilon)$ ("fast vanishing")

Without slow/critical vanishing (and previous assumptions), the pseudoinverse formula is valid.

Setting

$$\theta \in O \subseteq \Theta, \quad \min_{x \in \mathbb{R}^n} \left\{ f(x, \theta) + g(x, \theta) \quad \left(+ \frac{\varepsilon}{2} \|x\|_2^2 \right) \right\} \quad \text{(P-Comp)}$$

- everything is definable;
- $\forall\, \theta \in O$, $f(\cdot, \theta) \in \mathcal{C}^{1,1}(\mathbb{R}^n) \cap \Gamma_0(\mathbb{R}^n)$, $g(\cdot, \theta) \in \Gamma_0(\mathbb{R}^n)$;
- $\forall\, \theta \in O$, $\mathcal{S}_c(\theta) := \arg\min_{x \in \mathbb{R}^n} \{ f(x, \theta) + g(x, \theta) \} \neq \varnothing$;
- $\nabla_x f$ is jointly locally Lipschitz and verifies $\|\nabla_x f(x, \theta) - \nabla_x f(x', \theta)\| \leqslant L \|x - x'\|$;
- $G^\gamma(z, \theta) := \operatorname{prox}_{\gamma g(\cdot, \theta)}(z)$ is jointly locally Lipschitz ($\gamma > 0$).

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Composite Optimality Conditions

Fixed-point reformulation

$$x \text{ solution} \iff x - \underbrace{G^\gamma}_{\text{backward}} \Big(\underbrace{x - \gamma\big(\nabla_x f(x,\theta) + \varepsilon x\big)}_{\text{forward}}, \theta \Big) = 0$$

- conservative differentiation and nonsmooth IFT
- $J_{\nabla_x f} (\simeq [\partial_x \nabla_x f \ \ \partial_\theta \nabla_x f])$ and $J_{G^\gamma} (\simeq [\partial_x G^\gamma \ \ \partial_\theta G^\gamma])$
- complicated equations involving "$\partial \mathrm{prox}_{g(\cdot,\theta)} / \partial \theta$"

Requires stronger assumptions + one for "$\partial_\theta G^\gamma \simeq \partial \mathrm{prox}_{g(\cdot,\theta)} / \partial \theta$".

Still issues of $H_0^\dagger$ vs $\lim_{\varepsilon \to 0} \dots$ and which elements reached.

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